



NAME:
HABIB ID:
SECTION:

LINEAR ALGEBRA

SPRING 2026

QUIZ 6

Max Marks: 10

Time: 10 minutes

Q. Let $A = \begin{bmatrix} 2 & 4 & 2 \\ 2 & 6 & 8 \\ 4 & 14 & 22 \end{bmatrix}$. Find matrices L and U using LU decomposition of A .

Then verify your result by computing LU .



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Q. Let $A = \begin{bmatrix} 2 & 4 & 6 \\ 6 & 13 & 20 \\ 4 & 10 & 17 \end{bmatrix}$. Find matrices L and U using LU decomposition of A .

Then verify your result by computing LU .



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SOLUTIONS VERSION A

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 6 \\ 0 & 0 & 0 \end{pmatrix}$$

Part 1 — Compute L and U (8 pts)

- ☐ Correct first multipliers $m_{21} = 1$, $m_{31} = 2$ (2 pts)
- ☐ Correct second multiplier $m_{32} = 3$ (1 pt)
- ☐ U is upper triangular with correct entries (3 pts)
- ☐ L is unit lower triangular with correct entries (2 pts)

Part 2 — Verify $LU = A$ (2 pts)

- ☐ Matrix product set up correctly (1 pt)
- ☐ Final product equals A (1 pt)

VERSION B

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 4 & 6 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Part 1 — Compute L and U (8 pts)

- ☐ Correct first multipliers $m_{21} = 3$, $m_{31} = 2$ (2 pts)
- ☐ Correct second multiplier $m_{32} = 2$ (1 pt)
- ☐ U is upper triangular with correct entries (3 pts)
- ☐ L is unit lower triangular with correct entries (2 pts)

Part 2 — Verify $LU = A$ (2 pts)

- ☐ Matrix product set up correctly (1 pt)
- ☐ Final product equals A (1 pt)